For Examiner's Use

1 A student wishes to investigate how the resistance R of a light-dependent resistor varies with the distance d from an intense light source.

It is believed that the relationship between R and d is

$$R = kd^n$$

where k and n are constants.

Design a laboratory experiment to test the above relationship. The light-dependent resistor has a resistance of 100 Ω when it is in bright light and a resistance of 500 kΩ when no light falls on it.

You should draw a diagram showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements that would be taken,

(c) the control of variables,

(d) how the data would be analysed,

(e) any safety precautions that you would take.

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